



Faculté des Sciences et des Technologies

Laboratoire N° 6_ Réseaux II

Niveau : Licence III

Étudiant : Donsam Jean Gabard NOEL

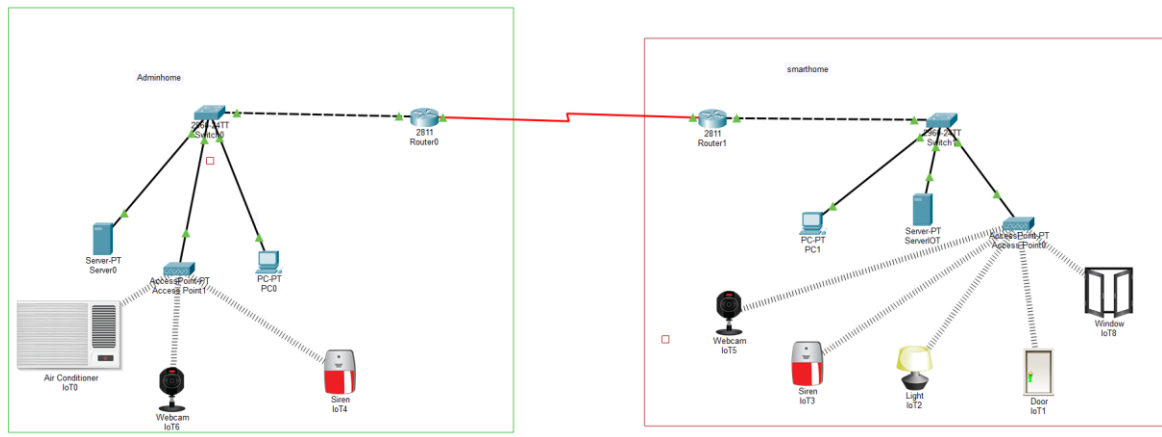
Professeur : Ismaël SAINT ARMOUR

Date : 3 Mai 2026

L'objectif de ce TD est de :

1. Le contrôle à distance
2. La communication entre objets
3. La sécurité du réseau IoT

Topologie



❖ Configuration de base des routeurs (R1 et R2)

Configuration de R1

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#interface Serial0/0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.252
R1(config-if)#clock rate 64000
This command applies only to DCE interfaces
R1(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
R1(config-if)#exit
R1(config)#interface FastEthernet0/1
R1(config-if)#ip address 192.168.2.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
exit
R1(config)#ip dhcp excluded-address 192.168.2.1 192.168.2.10
R1(config)#ip dhcp pool ADMINHOME
R1(dhcp-config)#network 192.168.2.0 255.255.255.0
R1(dhcp-config)#default-router 192.168.2.1
R1(dhcp-config)#dns-server 8.8.8.8
R1(dhcp-config)#exit
R1(config)#router ospf 1
R1(config-router)#network 192.168.1.0 0.0.0.3 area 0
R1(config-router)#network 192.168.2.0 0.0.0.255 area 0
R1(config-router)#exit
R1(config)#
```

Configuration de R2

```

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R2
R2(config)#interface Serial0/0/0
R2(config-if)#ip address 192.168.1.2 255.255.255.252
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
exit
R2(config)#interface F
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
astEthernet0/1
R2(config-if)#ip address 192.168.3.1 255.255.255.0
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
exit
R2(config)#ip dhcp excluded-address 192.168.3.1 192.168.3.10
R2(config)#ip dhcp pool SMARTHOM
R2(dhcp-config)#network 192.168.3.0 255.255.255.0
R2(dhcp-config)#default-router 192.168.3.1
R2(dhcp-config)#dns-server 8.8.8.8
R2(dhcp-config)#exit
R2(config)#router ospf 1
R2(config-router)#network 192.168.1.0
% Incomplete command.
R2(config-router)#network 192.168.1.0 0.0.0.3 area 0
R2(config-router)#network 192.168.1.0 0.0.0.255 area 0
00:15:20: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.2.1 on Serial0/0/0network 192.168.3.0 0.0.0.255 area 0
R2(config-router)#exit
R2(config)#

```

```

R2#ping 192.168.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/10/15 ms

R2#

```

Copy

Paste

❖ Configuration des Server IOT.

✓ Adminhome

Home Gateway0

Physical
Config
GUI
Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

Internet

LAN

Wireless

Wireless Settings

SSIDadminhome
2.4 GHz Channel6 - 2.437GHz
Coverage Range (meters)250,00

Authentication
☒ Disabled
☐ WEP
☐ WPA-PSK
☐ WPA2-PSK
☐ WPA
☐ WPA2

RADIUS Server Settings
IP Address
Shared Secret
Encryption TypeDisabled

Home Gateway0

Physical **Config** GUI Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

Internet

LAN

Wireless

LAN Settings

IP Configuration

IPv4 Address 192.168.2.3

Subnet Mask 255.255.255.0

Home Gateway0

Physical **Config** GUI Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

Internet

LAN

Wireless

Internet Settings

IP Configuration

☒ DHCP

☐ Static

IPv4 Address

Subnet Mask

Default Gateway

DNS Server

✓ Smarthome

Access Point0

Physical **Config** Attributes

GLOBAL

Settings

INTERFACE

Port 0

Port 1

Port 1

Port Status ☒ On

SSID smarthome

2.4 GHz Channel 6

Coverage Range (meters) 140,00

Authentication

☒ Disabled

☐ WEP

☐ WPA-PSK

☐ WPA2-PSK

WEP Key

PSK Pass Phrase

User ID

Password

Encryption Type Disabled

IoT0

SpecificationsPhysicalConfigAttributes

GLOBAL

Settings

Algorithm Settings

Files

INTERFACE

Wireless0

Bluetooth

Wireless0

Port Status

On

Bandwidth

300 Mbps

MAC Address

0009.7C9A.DB42

SSID

adminhome

Authentication

☒ Disabled

☐ WEP

WEP Key

☐ WPA-PSK

☐ WPA2-PSK

PSK Pass Phrase

☐ WPA

☐ WPA2

User ID

☐ 802.1X

Method:

Password

Encryption Type

Disabled

IP Configuration

☒ DHCP

☐ Static

IPv4 Address

192.168.25.111

Subnet Mask

255.255.255.0

IPv6 Configuration

☒ Automatic

☐ Static

IPv6 Address

Link Local Address

FE80::209:7CFF:FE9A:DB42

IoT0

SpecificationsPhysicalConfigAttributes

GLOBAL

Settings

Algorithm Settings

Files

INTERFACE

Wireless0

Bluetooth

☒ DHCP

☐ Static

Default Gateway

192.168.2.3

DNS Server

Gateway/DNS IPv6

☒ Automatic

☐ Static

Default Gateway

DNS Server

IoT Server

☐ None

☐ Home Gateway

☒ Remote Server

Server Address

192.168.2.2

User Name

admin

Password

admin123

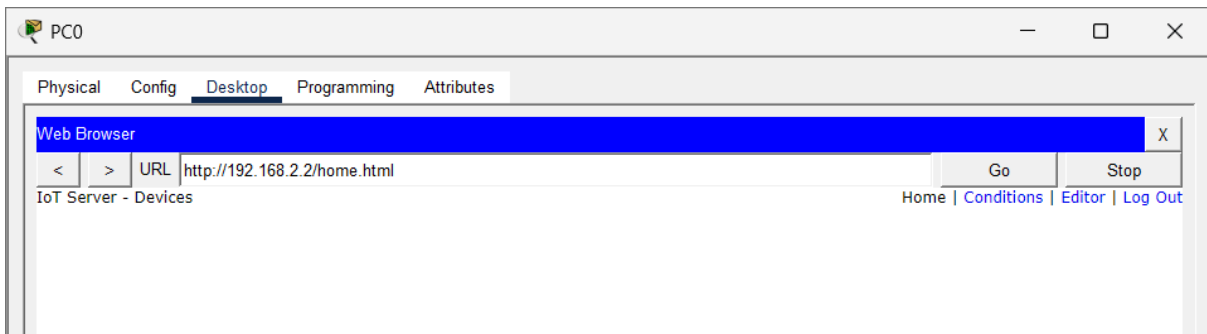
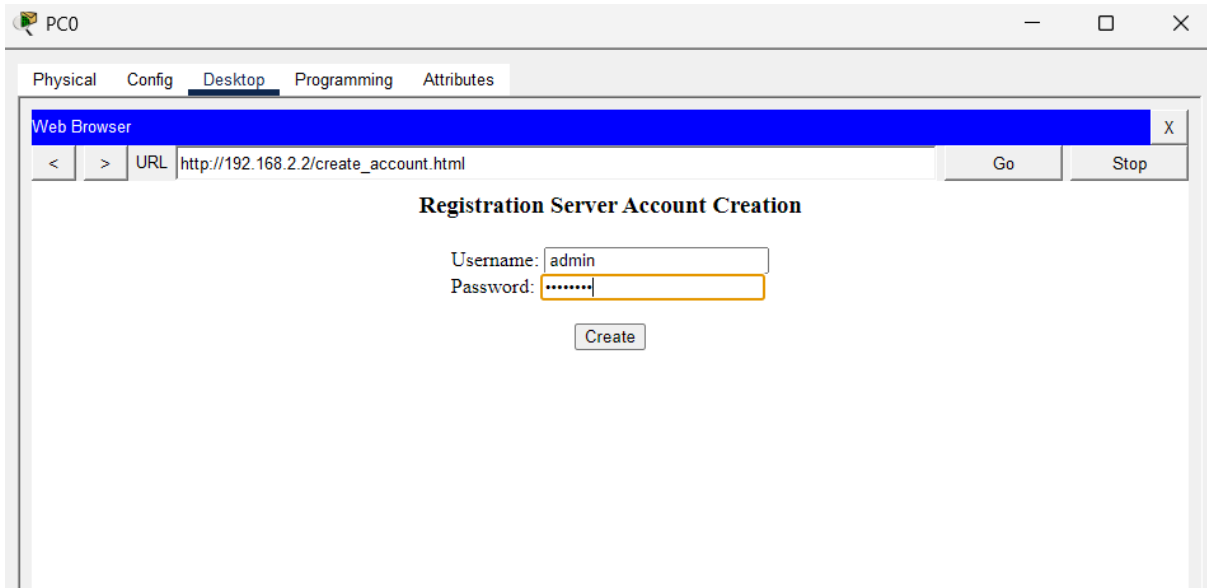
Connect

Device Clock:

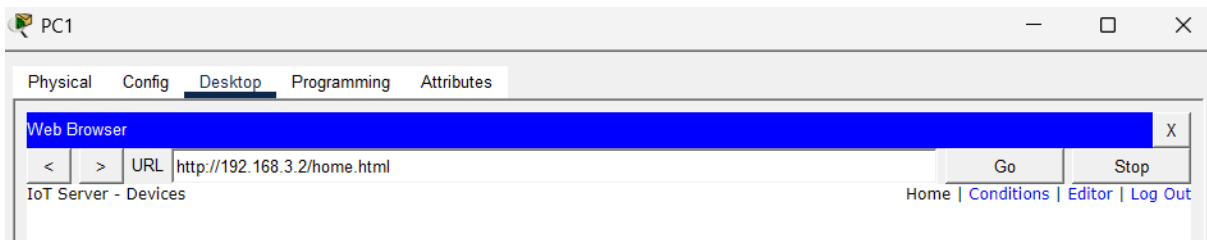
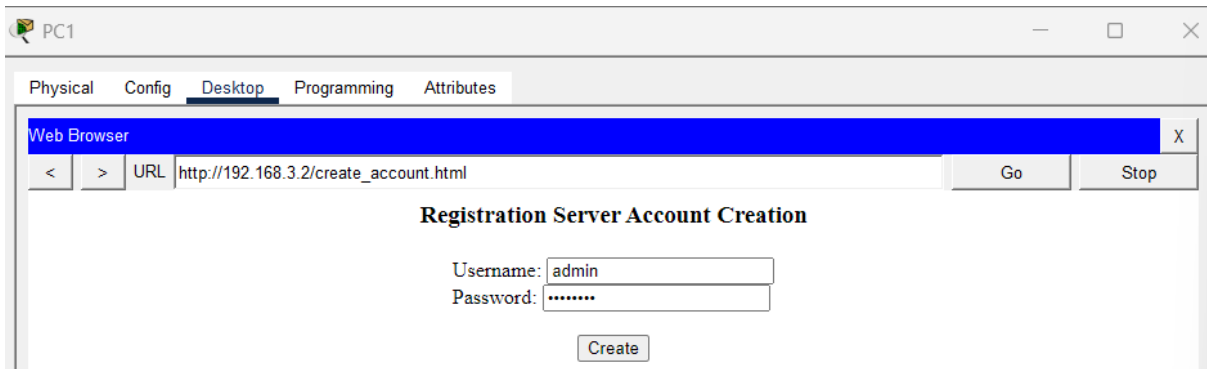
00:43:54 Mon Mar 1 1993 UTC

❖ Création du compte admin sur les serveurs IoT

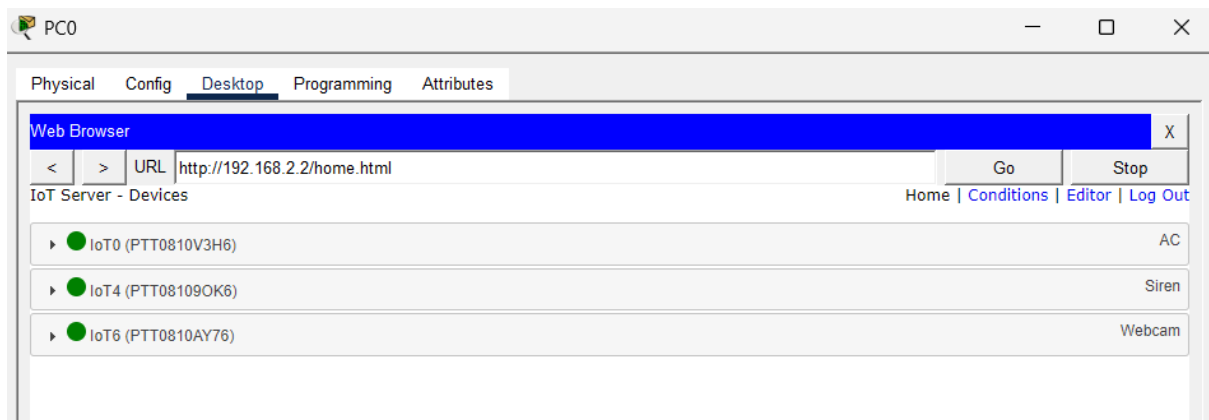
✓ Adminhome



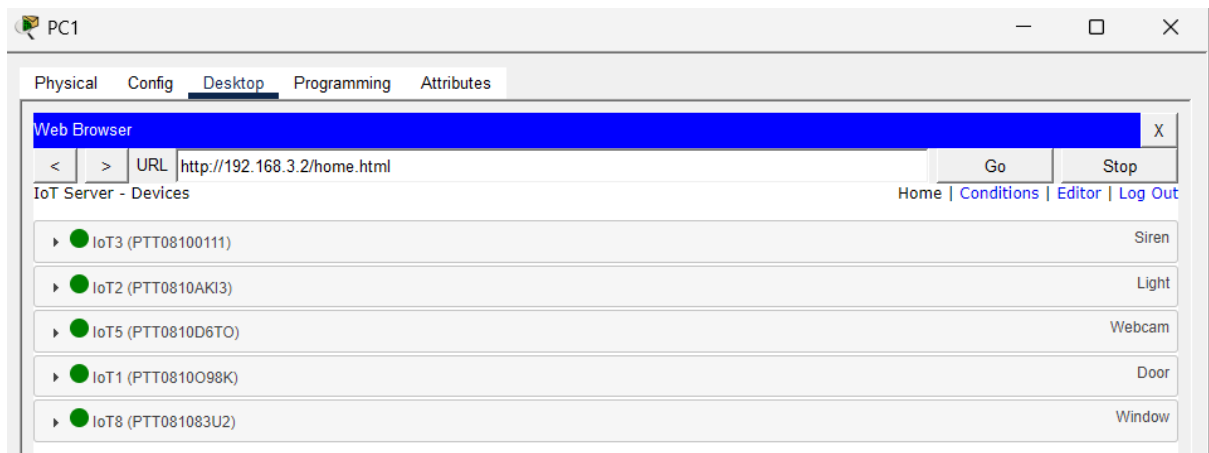
✓ Smarthome



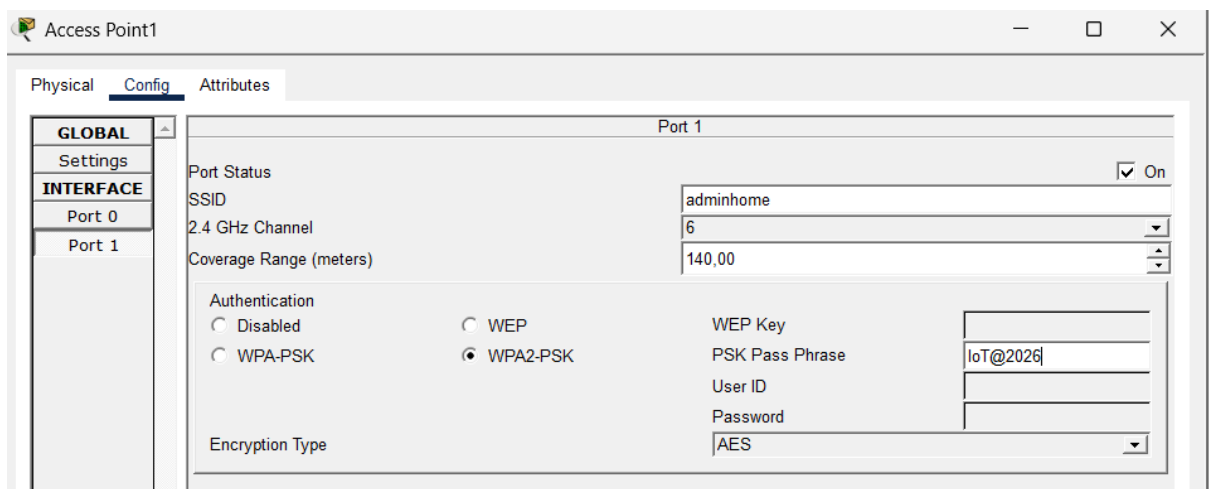
✓ Adminhome



✓ Smarthome



❖ Sécurité du réseau IoT



IoT0

Specifications Physical **Config** Attributes

GLOBAL

Settings

Algorithm Settings

Files

INTERFACE

Wireless0

Bluetooth

Wireless0

Port Status ☒ On

Bandwidth 18 Mbps

MAC Address 0009.7C9A.DB42

SSID adminhome

Authentication

☐ Disabled ☐ WEP ☒ WPA2-PSK

WEP Key

PSK Pass Phrase IoT@2026

❖ Automatisation IoT Rules

PC0

Physical Config **Desktop** Programming Attributes

Web Browser X

< > URL http://192.168.3.2/conditions.html Go Stop

IoT Server - Device Conditions [Home](#) | [Conditions](#) | [Editor](#) | [Log Out](#)

Actions	Enabled	Name	Condition	Actions
Add				

Add Rule X

Name rule1

Enabled ☒

If:

Match All

IoT1 Open is true

+ Condition + Group

Then set:

IoT5 On to true

IoT3 On to true

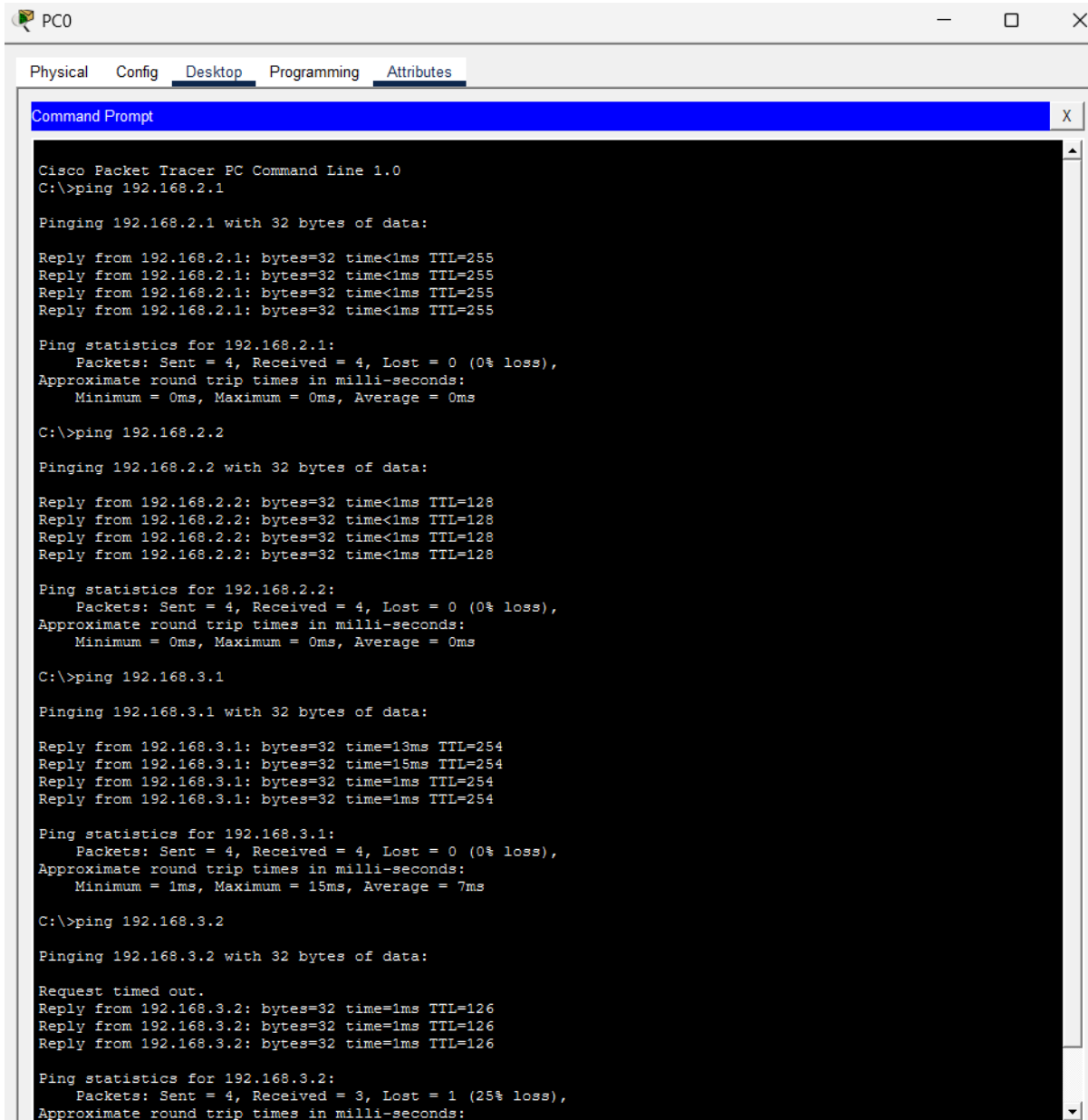
+ Action

OK Cancel

❖ Tests de connectivité

Depuis **PC0**

- ✓ ping 192.168.2.1 vers Router0
- ✓ ping 192.168.2.2 vers Server Io Adminhome
- ✓ ping 192.168.3.1 vers Router1 via OSPF
- ✓ ping 192.168.3.2 vers Server IoT Smarthome via OSPF



The screenshot shows the 'PC0' window in Cisco Packet Tracer. The 'Desktop' tab is active, displaying a 'Command Prompt' window. The command prompt shows the results of four ping commands executed from PC0. The first two pings (to 192.168.2.1 and 192.168.2.2) are successful with 0% loss. The third ping (to 192.168.3.1) is also successful with 0% loss. The fourth ping (to 192.168.3.2) shows a 'Request timed out' for the first packet, followed by three successful replies, resulting in a 25% loss.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.3.1

Pinging 192.168.3.1 with 32 bytes of data:

Reply from 192.168.3.1: bytes=32 time=13ms TTL=254
Reply from 192.168.3.1: bytes=32 time=15ms TTL=254
Reply from 192.168.3.1: bytes=32 time=1ms TTL=254
Reply from 192.168.3.1: bytes=32 time=1ms TTL=254

Ping statistics for 192.168.3.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 15ms, Average = 7ms

C:\>ping 192.168.3.2

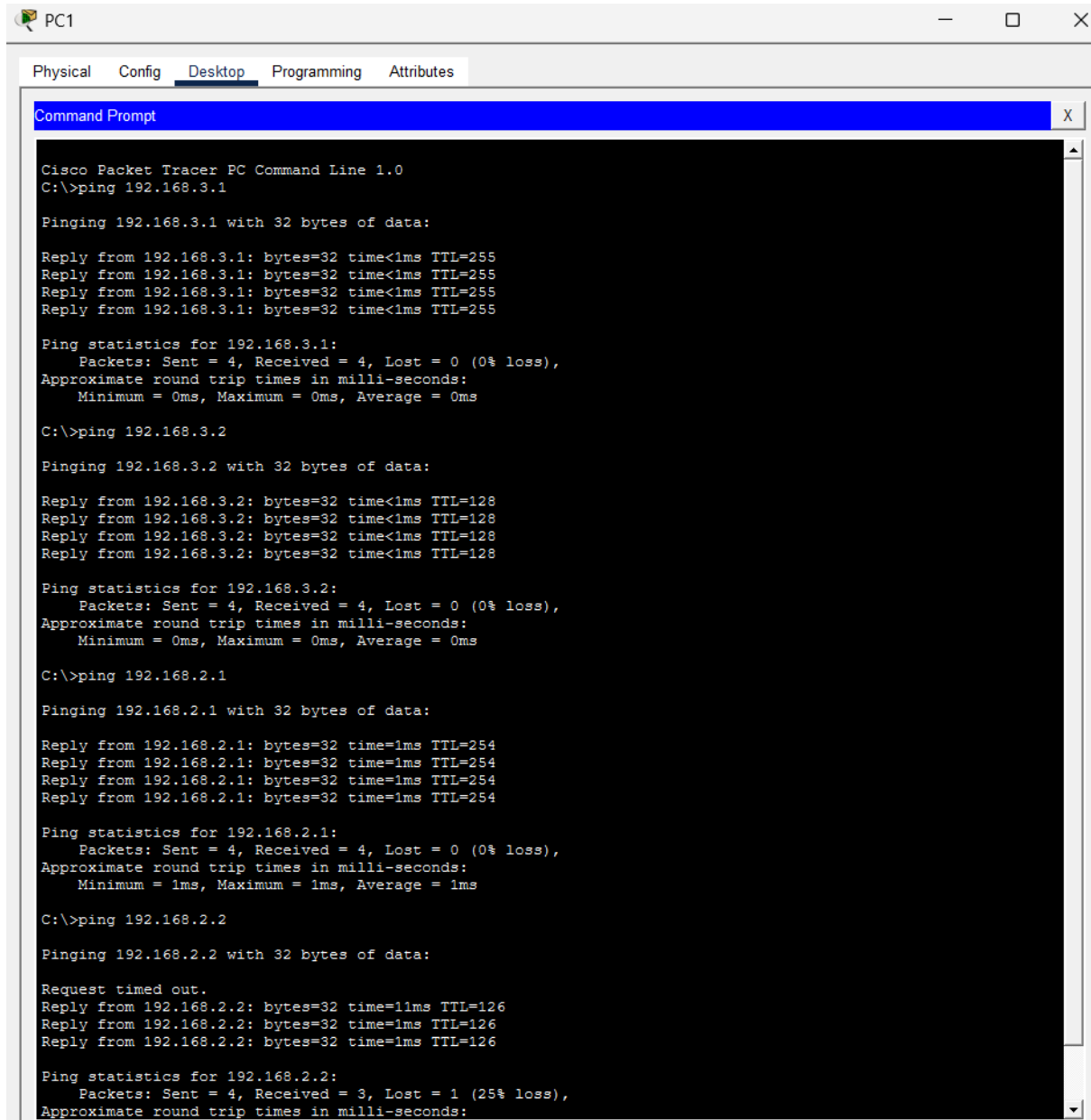
Pinging 192.168.3.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.2: bytes=32 time=1ms TTL=126
Reply from 192.168.3.2: bytes=32 time=1ms TTL=126
Reply from 192.168.3.2: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
```

Depuis PC1

- ✓ ping 192.168.3.1 vers Router1
- ✓ ping 192.168.3.2 vers Server IoT Smarthome
- ✓ ping 192.168.2.1 vers Router0 via OSPF
- ✓ ping 192.168.2.2 vers Server IoT Adminhome via OSPF



```
PC1
Physical  Config  Desktop  Programming  Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.1

Pinging 192.168.3.1 with 32 bytes of data:

Reply from 192.168.3.1: bytes=32 time<1ms TTL=255
Reply from 192.168.3.1: bytes=32 time<1ms TTL=255
Reply from 192.168.3.1: bytes=32 time<1ms TTL=255
Reply from 192.168.3.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.3.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Reply from 192.168.3.2: bytes=32 time<1ms TTL=128
Reply from 192.168.3.2: bytes=32 time<1ms TTL=128
Reply from 192.168.3.2: bytes=32 time<1ms TTL=128
Reply from 192.168.3.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Reply from 192.168.2.1: bytes=32 time=1ms TTL=254
Reply from 192.168.2.1: bytes=32 time=1ms TTL=254
Reply from 192.168.2.1: bytes=32 time=1ms TTL=254
Reply from 192.168.2.1: bytes=32 time=1ms TTL=254

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.2: bytes=32 time=11ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
```

✓ Test contrôle à distance des objets IoT

